



Vehicle Trajectory Prediction based on LSTM Recurrent Neural Networks

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Summary

- **Trajectory Prediction:** LSTM RNN model proposed for vehicle trajectory prediction using real taxi data, addressing challenges in existing trajectory prediction models and exploring impact of prior data
- **Architecture:** LSTM utilized to mitigate vanishing gradient problem in long-term sequential data, with GPS coordinates converted to grid map cells and trajectories partitioned into sequences
- **Results:** Improved prediction performance (over 89% prediction accuracy) with more prior observations, but increased computation time
- **Keywords:** Trajectory Prediction, LSTM, RNN

Abstract

- A Long Short-Term Memory (LSTM) Recurrent Neural Network (RNN) scheme for predicting vehicle trajectories based on current and previous locations is proposed, utilizing real taxi data from Porto, Portugal
- Demonstrates improved prediction with more prior mobility information and achieves over 89% prediction performance

Introduction

- Vehicle trajectory prediction is crucial for various applications such as mobility management, data traffic optimization, and quality of service (QoS) improvement in vehicular and wireless networks
- Existing trajectory prediction models are categorized into short-term and long-term models, with short-term models being more common due to the challenges posed by the randomness of vehicle positions over time
- The LSTM based neural network proposed for predicting taxi trajectories in urban areas, aims to address these challenges and explore the impact of different amounts of prior data

Literature Review and Contributions

- Existing studies on trajectory prediction include methods like dynamic Bayesian networks and LSTM-based models, each with their limitations such as computational intensity or applicability to specific scenarios
- This paper introduces a LSTM neural network designed for predicting vehicle trajectories using prior sequence data, along with preprocessing of raw trajectory data for training
- It evaluates the impact of varying amounts of prior data on prediction performance and computation time, demonstrating high prediction performance and low computation time for the proposed model

Preliminaries

- The study focuses on analyzing vehicle trajectories in a specific region represented by a two-dimensional grid map, where each geographical sub-region is denoted as a cell

- Trajectories are represented by sequences of visited cells, with each trajectory containing a variable number of sequentially visited cells
- Sequences are formed by consecutive visited cells and are considered as units of inference during the network training phase, with the set of sequences containing both total and unique sequences
- The dataset used consists of real traces from 442 taxis in Porto, Portugal, collected over a year, with relevant features including trip ID, timestamp, and GPS coordinates
- Data preprocessing involves converting the GPS coordinates to grid map cells and partitioning trajectories into sequences

LSTM for Trajectory Prediction

- The LSTM structure addresses the vanishing gradient problem commonly encountered in dealing with long-term sequential data, sharing cell states across each forward step
- The LSTM layer consists of $\Lambda-1$ discrete time steps, utilizing the $\Lambda-1$ prior observed cells to predict the next one, with input vectors X_i containing information about the visited cell represented using one-hot encoding
- In each LSTM layer, 16 LSTM units were adopted, and the final Dense Layer uses a sigmoid activation function to produce the final one-hot-encoded output
- The network is trained using a dataset containing sequences of visited cells, with 70% used for training and 20% for validation, employing the Adam optimizer and categorical cross entropy as the loss function, with early stoppage algorithm implemented to avoid overfitting

Experiment Results

- The LSTM network was evaluated using Python, Tensorflow, and Keras, and trained on a NVIDIA GeForce RTX-2080 GPU and an Intel Core i7-8750H 2.2 GHz processor
- Prediction performance is assessed by comparing predicted cells with true cells from the test set, with the ratio of correctly predicted cells over the total

number of sequences determining prediction performance

- Results show that prediction performance improves with an increase in Λ , indicating better performance as more prior observations are considered in the prediction process
- Confusion matrices generated for $\Lambda = 4$ and $\Lambda = 20$ illustrate high prediction performance ratios, with the $\Lambda = 20$ method achieving higher performance for almost all cells. Additionally, computational efficiency analysis reveals that computation time increases linearly with the number of sequences for all LSTM prediction methods

Conclusion

- Proposed LSTM RNN for accurate and efficient vehicle trajectory prediction
- Experimental results show prediction performance exceeding 89% and improved accuracy with more prior observations